

Gaurav Singh

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Research profile

Applied machine-learning researcher with publications on visual-mathematical literacy, deformation detection in auxetic structures, and transport decarbonisation modelling. Former head of the Master of Design in Design Computation and founder of the Experimental Maths Lab at Srishti Manipal Institute. Maintain Mathscales as an independent mathematical practice.

Education

2023-2025	Master of Information Technology Griffith University, Gold Coast, Australia; GPA 6.31/7 Thesis: "Enhancing Climate Forecasting with Advanced Generative Models for Synthetic AQI Data", supervised by Dr. Elizabeth Chang.
2013-2015	Advanced Diploma, Innovation & Experience Design <u>Srishti Institute of Art, Design and Technology, Bangalore, India</u>
2008-2012	B.Tech, Computer Science & Engineering JNTU Anantapur, India; First Class with Distinction

Publications

A Framework for Visual-Mathematical Literacy in Applied Machine Learning: Why Representation Choices Shape What Models Can Learn. G. Singh, R. S. Dhari. Machine Learning: Engineering, 2026. <https://doi.org/10.1088/3049-4761/ae7df3>

Techno-Economic Pathways Modeling and Nonlinear Optimized SEEA-ROI Longitudinal Dynamic Simulation for Decarbonizing Australian Heavy Transportation Systems. G. Singh, E. Chang, Y. Karaca. Fractals, 2026. <https://doi.org/10.1142/S0218348X26400633>

Automated detection of deformation mechanisms in re-entrant honeycomb auxetics using machine learning. G. Singh, R. S. Dhari, Z. Javanbakht. International Journal of Protective Structures, 2025. <https://doi.org/10.1177/20414196241281069>

ReRide: A Bike Area Network for Embodied Self-monitoring during Motorbike Commute. N. Bagalkot, G. Singh, V. Rath, T. Sokoler, A. Shukla. TEI 2019, 443-450, 2019. <https://doi.org/10.1145/3294109.3300986>

ReRide. N. Bagalkot, T. Sokoler, R. Shaikh, G. Singh, A. E. Lillie, P. Dixit, A. Rai, V. Chakravarthy, A. Senthil. INTERACT 2017, LNCS 10516, 2017. https://doi.org/10.1007/978-3-319-68059-0_43

SnapTag: Leveraging Situated Memory to enhance self-efficacy for well-being. S. Baadkar, G. Singh, A. Saraf, N. Bagalkot. India HCI 2014, 136-141, 2014. <https://doi.org/10.1145/2676702.2676719>

Academic appointment and leadership

- 2017-2022 Faculty Member and Researcher, Srishti Manipal Institute, Bangalore, India
- Headed the Master of Design in Design Computation.
 - Founded the Experimental Maths Lab.
 - Taught programming, algorithms, machine learning, physical computing, and interaction design.

Teaching and curriculum

Designed and taught 30 courses and workshops (2013-2022), primarily at Srishti Manipal Institute in Bangalore, spanning creative coding, algorithmic botany, physical computing, machine learning, interaction design, and human-computer interaction.

Designed and facilitated Deconstructing Algorithms, a 2020 undergraduate thesis studio with five students investigating algorithmic literacy through explainable AI and impact-investment information.

Technical methods

Python, scikit-learn, computer vision, time-series and generative modelling, statistical inference, nonlinear optimisation, experimental design, and data visualisation.

Thesis supervision

- 2020 Discovering Algorithmic Literacy through Explainable AI
Thesis supervisor; students: D. Dixit, S. Singh
- 2020 Financial indicators that motivate impact investment
Thesis supervisor; students: A. Avadhana, N. Patny, S. Mishra
- 2013 Location and arrival prediction
Thesis supervisor; students: Shobha R., Pavani A., Kalpana, JNTU Anantapur

Academic service

- 2018 Demonstrations track co-chair, IndiaHCI 2018
Bangalore, India
- 2017 Cumulus Conference, organising team
Bangalore, India

Awards

- 2024 Griffith Award for Academic Excellence (Griffith University)
- 2024 STEM Professional Gold Achievement (Griffith University)
- 2017 Topcoder Open Design Champion (India Regionals)
- 2012 Topcoder Open Studio Finals (6th, Orlando)
- 2011 Topcoder Open Studio Finals (11th, Fort Lauderdale)

Grants

- 2023 Adobe Fund for Design (Design grant)
- 2021 Adobe Fund for Design (Design grant)

Website: <https://gaurav-singh.info> · GitHub: github.com/gv-sh